

Dylan Costa

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RESEARCH INTERESTS

Algebraic Number Theory and Arithmetic Geometry, with a focus on elliptic curves, modular curves, and Galois representations attached to abelian varieties.

EDUCATION

University of Connecticut PhD in Mathematics Advisor: Álvaro Lozano-Robledo	2021 – 2027 (<i>expected</i>)
Ithaca College BA in Mathematics, <i>with Honors</i> Advisors: Emilie Wiesner & Ted Galanthay	2017 – 2021

TEACHING EXPERIENCE

Instructor of Record

MATH 2410: Elementary Differential Equations <i>2 sections of 40 students each semester</i>	Fall 2025, Spring 2026
MATH 1071: Calculus for Business and Economics <i>2 sections of 40 students each semester</i>	Fall 2023, Spring 2024, Spring 2025

Graduate Teaching Assistant

MATH 3160: Probability (<i>Online</i>)	Summer 2025, Summer 2026
MATH 2110: Multivariable Calculus <i>3 sections of 40 students each semester</i>	Summer 2024, Fall 2024
MATH 1132: Calculus II <i>3 sections of 40 students each semester</i>	Spring 2022, Summer 2022, Fall 2022, Spring 2023, Summer 2023, Summer 2026
MATH 1131: Calculus I <i>2 sections of 40 students each semester</i>	Fall 2026, Fall 2021

MENTORING

Independent Study: Cryptography <i>University of Connecticut</i>	Spring 2026
Directed Reading Project in Algebraic Number Theory <i>University of Connecticut Directed Reading Program</i>	Fall 2025
Directed Reading Project on Algebraic Curves <i>University of Connecticut Directed Reading Program</i>	Spring 2024

PUBLICATIONS

<i>ℓ-primary Torsion in Elliptic Curves over $\mathbb{Q}$</i> With Álvaro Lozano-Robledo and Enrique González-Jiménez.	In preparation
<i>Classification of Isogeny-Torsion Graphs in Quadratic Fields</i> With Rakvi.	In preparation

INVITED TALKS

Recent Discoveries on Ranks of Elliptic Curves <i>SIGMA Graduate Seminar, University of Connecticut</i>	October 2026
A Computational Perspective on Modular Curves <i>Number Theory Reading Group, University of Connecticut</i>	November 2025
Rank Growth of Elliptic Curves over Quadratic Extensions <i>SIGMA Graduate Seminar, University of Connecticut</i>	August 2025
Elliptic Curves and their Applications in Mathematics and Beyond <i>Mathematics Colloquium, Ithaca College</i>	April 2025
Pontryagin Duality and Applications to Tate’s Thesis <i>Number Theory Reading Group, University of Connecticut</i>	March 2025
Rank Growth of Elliptic Curves over Quadratic Extensions <i>Graduate Research Forum, University of Connecticut</i>	January 2025
On ℓ-adic Images of Galois Attached to Elliptic Curves over the Rationals <i>General Exam Presentation, University of Connecticut</i>	November 2024
On the Euclidean Embedding and Adeles/Ideles of a Number Field <i>Number Theory Reading Group, University of Connecticut</i>	October 2024
Construction of High Rank Elliptic Curves over Quadratic Fields <i>Number Theory Reading Group, University of Connecticut</i>	April 2024
Elliptic Curves and their Applications <i>Mathematics Continued Conference, University of Connecticut</i>	March 2024
On Meromorphic Differentials and Riemann-Roch <i>Number Theory Reading Group, University of Connecticut</i>	April 2023

AWARDS

Summer Travel Fellowship <i>University of Connecticut</i>	Summer 2026
Oracle Honors Society <i>Ithaca College</i>	Fall 2019 – Spring 2021
Summer Scholar Fellowship <i>Ithaca College</i>	Summer 2020
Ramanujan Award in Mathematics <i>Ithaca College</i>	Fall 2019

RELEVANT SKILLS

Programming	MAGMA, SAGE, Java (<i>advanced</i>), Python, MATLAB (<i>intermediate</i>), HTML (<i>novice</i>)
Software	L ^A T _E X, Excel, Google Sheets